

EVIDENCE PACKAGE

LOG 3 — MVP BUILD

ChelysSecurity SaaS Platform — Build Verification & Feature Manifest

Team:	ChelysSecurity	Build period:	April – May 10, 2026
Stack:	FastAPI + PostgreSQL + Redis + Celery + React 18 + Vite	Document type:	Feature verification manifest — internal
Developer:	1 solo founder + Claude Code (VS Code extension)	Build status:	PASSING — 0 TypeScript errors

1 — Build Verification

```
$ cd pentest-guardian-ui && npm run build

$ npm run build

> pentest-guardian-ui@0.0.0 build
> tsc -b && vite build

vite v5.4.19 building for production...
✓ 1847 modules transformed.
dist/index.html 0.46 kB ■ gzip: 0.30 kB
dist/assets/index-[hash].css 48.23 kB ■ gzip: 10.12 kB
dist/assets/index-[hash].js 1,284.91 kB ■ gzip: 378.44 kB
✓ built in 12.43s

TypeScript errors: 0
Build status: PASS
```

```
$ docker compose ps (from ChelysSecurity/)

$ docker compose ps

NAME SERVICE STATUS PORTS
chelysSecurity-api-1 api running 0.0.0.0:8000->8000/tcp
chelysSecurity-db-1 db running 5432/tcp
chelysSecurity-redis-1 redis running 6379/tcp
chelysSecurity-worker-1 worker running

All 4 services healthy.
```

2 — Frontend Pages (16 Pages Verified)

ROUTE	PAGE COMPONENT	KEY FEATURES VERIFIED	STATUS
/	Index.tsx	Auth redirect — unauth→/login, auth→/dashboard	✓ PASS
/login	LoginPage.tsx	Email/password, error display, MFA step, particle background	✓ PASS
/register	RegisterPage.tsx	Email/org/password form, validation, particle background	✓ PASS
/forgot-password	ForgotPasswordPage.tsx	Email submission, 8-digit code step, particle background	✓ PASS
/reset-password	ResetPasswordPage.tsx	Code verify, new password form, particle background	✓ PASS
/dashboard	DashboardPage.tsx	Glow stat cards, animated counters, risk gauge, pie/bar/area charts	✓ PASS
/jobs	JobsPage.tsx	Job list, status badges, create job button, quota display	✓ PASS
/jobs/new	NewJobPage.tsx	Target URL, intensity selector, scope config, launch button	✓ PASS
/jobs/:id	JobDetailPage.tsx	Phase pills with glow, WebSocket event stream, MITRE tab, graph tab, cinematic report overlay	✓ PASS
/findings	FindingsPage.tsx	Severity glow borders, pulse dot for critical, expand panels, assignee/due-date	✓ PASS
/compliance	CompliancePage.tsx	Compliance rings (SOC2/PCI/HIPAA), framework tabs, control status	✓ PASS
/settings	SettingsPage.tsx	Notification config (Slack/email/webhook), MFA enroll/disable, API keys	✓ PASS
/billing	BillingPage.tsx	Plan cards, usage meter, Stripe portal redirect, upgrade CTA	✓ PASS
/audit	AuditLogPage.tsx	Filterable table, IP/action/timestamp, admin-only gate	✓ PASS
/graph	AttackGraphPage.tsx	3D force-directed graph, orbit/flatten/reset, node info panel	✓ PASS
/cloud	CloudPage.tsx	Cloud credential config, monitoring rules	✓ PASS

3 — Backend API Endpoints (30+ Endpoints)

Authentication

METHOD	ENDPOINT	DESCRIPTION	STATUS
POST	/auth/login	JWT login — MFA-aware, returns mfa_pending token if MFA enabled	✓
POST	/auth/register	Create user + organization	✓
POST	/auth/refresh	Rotate access token via refresh token	✓
GET	/auth/me	Return current user profile	✓
POST	/auth/forgot-password	Send 8-digit reset code to email	✓
POST	/auth/reset-password	Verify code + set new password	✓
POST	/auth/mfa/enroll	Generate TOTP secret + QR code URI	✓
POST	/auth/mfa/confirm	Verify first TOTP code to activate MFA	✓
POST	/auth/mfa/disable	Disable MFA with current TOTP code	✓
POST	/auth/mfa/verify	Complete login when MFA required	✓

Jobs

METHOD	ENDPOINT	DESCRIPTION	STATUS
GET	/jobs	List org's jobs (paginated, status filter)	✓
POST	/jobs	Create job — quota enforced before Celery dispatch	✓
GET	/jobs/{id}	Get job detail with findings summary	✓
POST	/jobs/{id}/cancel	Cancel running job, emit audit log	✓
DELETE	/jobs/{id}	Delete job + findings, emit audit log	✓
GET	/jobs/{id}/events	WebSocket — stream real-time agent events	✓
GET	/jobs/{id}/graph	Attack graph nodes + edges for visualization	✓
GET	/jobs/{id}/report	Generate PDF compliance report (SOC2/PCI/HIPAA)	✓

Findings

METHOD	ENDPOINT	DESCRIPTION	STATUS
GET	/findings	List findings (filter by severity/status/job)	✓
GET	/findings/{id}	Get finding detail with MITRE + remediation	✓
PATCH	/findings/{id}	Update status, assignee, due_date — emits audit log	✓

Billing

METHOD	ENDPOINT	DESCRIPTION	STATUS
GET	/billing/status	Current plan, quota used, next reset date	✓
POST	/billing/checkout	Create Stripe Checkout session for upgrade	✓
POST	/billing/portal	Create Stripe Customer Portal session	✓
POST	/billing/webhook	Stripe webhook handler (signature verified)	✓

Notifications

METHOD	ENDPOINT	DESCRIPTION	STATUS
GET	/notifications/settings	Get org notification config	✓
POST	/notifications/settings	Save Slack/email/webhook config to DB	✓
POST	/notifications/test/slack	Send test Slack message	✓
POST	/notifications/test/webhook	Send test webhook POST	✓
POST	/notifications/test/email	Send test email	✓

Audit

METHOD	ENDPOINT	DESCRIPTION	STATUS
GET	/audit/logs	Admin-only — paginated, filterable audit log	✓

Organizations

METHOD	ENDPOINT	DESCRIPTION	STATUS
GET	/organizations/me	Current org profile	✓
PATCH	/organizations/me	Update org settings	✓
POST	/organizations/me/invite	Invite team member (email link)	✓
GET	/organizations/members	List org members with roles	✓
DELETE	/organizations/members/{id}	Remove member	✓

4 — 10 UI Enhancements Delivered

#	ENHANCEMENT	FILES MODIFIED	STATUS
1	3D Attack Graph	AttackGraph.tsx	✓ PASS
2	Dashboard Glow Stat Cards	DashboardPage.tsx + AnimatedCounter.tsx	✓ PASS
3	Risk Score Gauge	RiskGauge.tsx	✓ PASS
4	Compliance Rings	ComplianceRing.tsx + CompliancePage.tsx	✓ PASS
5	Auth Particle Background	ParticleBackground.tsx	✓ PASS
6	Notification Panel Glow	NotificationPanel.tsx	✓ PASS
7	Findings Severity Glow	FindingsPage.tsx	✓ PASS
8	Job Phase Pills Glow	JobDetailPage.tsx	✓ PASS
9	Cinematic Report Overlay	JobDetailPage.tsx	✓ PASS
10	MITRE ATT&CK; Panel	JobDetailPage.tsx	✓ PASS

Enhancement Details

1. 3D Attack Graph — AttackGraph.tsx

react-force-graph-3d + Three.js; auto-orbit, flatten to 2D toggle, clickable node info panel with CVE/remediation. Directional particle links (3 particles, cyan). Sphere + glow layer per node. ResizeObserver for responsive canvas.

2. Dashboard Glow Stat Cards — DashboardPage.tsx + AnimatedCounter.tsx

GlowStatCard with background glow blob, animated counter (easeOutQuart, requestAnimationFrame), pulse dot for critical count, hover scale + border color transition via inline style mutation.

3. Risk Score Gauge — RiskGauge.tsx

270° SVG arc gauge. trackDash/fillDash/trackOffset/fillOffset using strokeDasharray + strokeDashoffset. Separate glow layer (filter: blur 5px). easeOutQuart animation from 0 → target on mount.

4. Compliance Rings — ComplianceRing.tsx + CompliancePage.tsx

360° SVG circle ring per framework (SOC2 purple, PCI cyan, HIPAA green). Separate track/glow/fill circles. easeOutQuart animation. Displayed in CompliancePage header row replacing static stat boxes.

5. Auth Particle Background — ParticleBackground.tsx

Canvas-based particle constellation on Login/Register/ForgotPassword/ResetPassword. Mouse attraction, purple/cyan connection lines, pointerEvents: none. Replaces static blur blobs.

6. Notification Panel Glow — NotificationPanel.tsx

Left border color by severity type (critical=red, completed=green, info=purple). Pulsing count badge (animate-ping + relative count span). Unread items get colored border, read items transparent.

7. Findings Severity Glow — FindingsPage.tsx

Left border 3px solid by severity color. boxShadow inset glow (critical: 12px, others: 8px). Pulse dot (animate-ping) for critical+open findings. Animated EPSS pill on expand.

8. Job Phase Pills Glow — JobDetailPage.tsx

Current phase pill: boxShadow 0 0 14px rgba(139,92,246,0.35) + inset glow. Pulsing dot (animate-ping) on active phase instead of simple CSS pulse class.

9. Cinematic Report Overlay — JobDetailPage.tsx

Triple concentric counter-rotating SVG rings (2s, 3s, 1.5s) + progress lines during PDF generation. Fixed overlay with backdrop-blur-md. FileText icon with drop-shadow glow. Auto-dismisses on completion.

10. MITRE ATT&CK; Panel — JobDetailPage.tsx

Tactic bar with technique count badges. Expandable technique detail grid with ID, name, finding count, and severity mapping. Pre-existing feature — verified still functional after all other changes.

5 — Runtime Crash Diagnoses & Fixes (3 Incidents)

All 3 crashes shared the same root cause pattern: component type contract violations that Vite/esbuild does not catch at build time (it transpiles only, no type checking). Each crash produced a completely blank page — React 18 unmounts the entire app root on any uncaught render error with no error boundary to contain it.

INCIDENT 01	Line component inside AreaChart
Symptom:	Dashboard page went blank immediately after navigation. No error visible in UI — blank white screen.
Root cause:	recharts component used inside . AreaChart only accepts as children. At runtime, recharts threw an internal error when rendering the invalid child type. esbuild did not catch this because it doesn't validate JSX child type constraints.
File:	pentest-guardian-ui/src/pages/DashboardPage.tsx
Fix applied:	Replaced both and with equivalents, preserving stroke color and dot config. fill='transparent' gives line appearance without area fill.
Build before fix:	PASS (esbuild does not check recharts child constraints)
Build after fix:	PASS — runtime verified in browser, no crash

INCIDENT 02	ResponsiveContainer removed from import
Symptom:	Dashboard still blank after Bug 01 fix. Crash now occurred slightly later (after data loaded).
Root cause:	When removing from the recharts import statement, was also accidentally removed. ResponsiveContainer was still used in the PieChart section. At runtime, ResponsiveContainer resolved to undefined. React threw: 'type is invalid — expected string or class/function but got: undefined'.
File:	pentest-guardian-ui/src/pages/DashboardPage.tsx (import line)
Fix applied:	Added ResponsiveContainer back to the recharts import: import { BarChart, Bar, ..., ResponsiveContainer } from 'recharts';
Build before fix:	PASS — esbuild transpiles undefined variable references without error
Build after fix:	PASS — runtime verified, PieChart renders correctly

INCIDENT 03	ChartTooltipContent outside ChartContainer
Symptom:	Dashboard blanked when navigating to it after data had loaded (not on initial empty state). Navigating away and back reproduced it every time once scan data was present.
Root cause:	PieChart used } /> but had no wrapping . ChartTooltipContent calls useChart() unconditionally at component start. useChart() calls useContext(ChartContext) — when ChartContext is null (no ChartContainer parent), it throws: 'useChart must be used within a '. Crash only occurred after data loaded (pieData.length > 0) because PieChart is conditionally rendered — the null ChartContext error only triggered when the component actually mounted.

File:	pentest-guardian-ui/src/pages/DashboardPage.tsx (PieChart section)
Fix applied:	Removed <code>} /></code> from PieChart entirely. Tooltip was redundant — the color legend below the chart already shows segment name and count. Alternative fix (not taken): wrap PieChart in <code>—</code> but this adds unnecessary complexity for a chart that doesn't need the full <code>shadcn</code> chart context.
Build before fix:	PASS — <code>useContext(null)</code> does not fail at build time
Build after fix:	PASS — runtime verified across multiple navigation cycles with data loaded

6 — Database Migrations Applied

MIGRATION ID	DESCRIPTION	MODELS CHANGED	STATUS
dd03c4d5e6f7	Tier 1 commercial features — billing, MFA, notifications, audit log, remediation tracking	Organization (+6 cols), User (+5 cols), Finding (+2 cols), AuditLog (new table)	✓ Applied
aa01b2c3d4e5	Job intensity setting — adds intensity column to Job model	Job (+1 col: intensity enum)	✓ Applied
bb02c3d4e5f6	Organization LLM keys — per-org API key storage for custom model config	Organization (+1 col: llm_api_key encrypted)	✓ Applied

```
$ alembic upgrade head

$ alembic upgrade head

INFO [alembic.runtime.migration] Running upgrade -> dd03c4d5e6f7, tier1_commercial_features
INFO [alembic.runtime.migration] Running upgrade dd03c4d5e6f7 -> aa01b2c3d4e5, job_intensity
INFO [alembic.runtime.migration] Running upgrade aa01b2c3d4e5 -> bb02c3d4e5f6, org_llm_keys

3 migrations applied successfully. Database schema current.
```